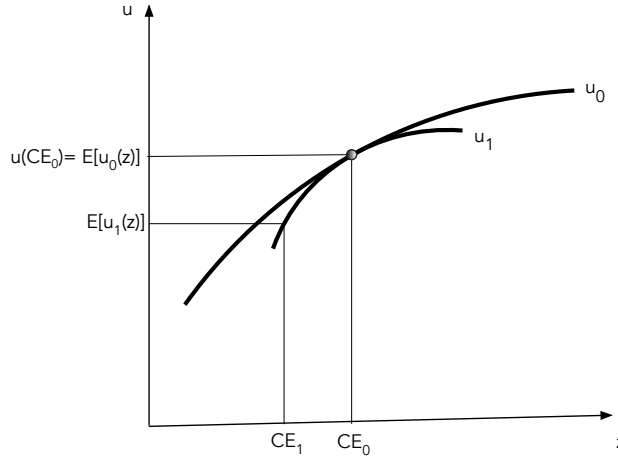


Comparative Risk Aversion (The Arrow-Pratt Theorem)

PROOF SKETCH THAT “MORE CONCAVE THAN” IMPLIES A HIGHER RISK PREMIUM (“2 implies 4” in my numbering). Let $u_1(z) = T(u_0(z))$ for every $z \geq 0$, where $T : \text{Range}(u_0) \rightarrow \mathbb{R}$ is concave and u_t , $t = 0, 1$ is C^2 with $u' > 0$. Fix a risk F , and let CE_t , $t = 0, 1$ be the certainty equivalent for this risk for preferences represented by the vN-M utility u_t . Without loss of generality we can suppose that $u_1(CE_0) = u_0(CE_0)$ and $u'_1(CE_0) = u'_0(CE_0)$. This follows since we can set $\tilde{u}_1 = a + bu_1$, where $\tilde{u}'_1(CE_0) = bu'_0(CE_0)$ and $\tilde{u}_1(CE_0) = a + bu_0(CE_0)$. In what follows omit the “tilde.” Since $u'_1(z) = T'(u_0(z))u'_0(z)$



and T is concave, u_1 is flatter than u_0 to the right of $z = CE_0$ and steeper than u_0 to the left, as in the Figure. In this case note that $\int u_1(z)dF(z) \leq \int u_0(z)dF(z)$. Recall the normalization $u_0(CE_0) = u_1(CE_0)$ to conclude that

$$u_1(CE_1) = \int u_1(z)dF(z) \leq \int u_0(z)dF(z) = u_0(CE_0) = u_1(CE_0).$$

Since u_1 is strictly increasing, $CE_1 \leq CE_0$, and therefore that $RP_1 \geq RP_0$.